

Oligodendrocytes as regulators of neuronal networks during early postnatal development.

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Oligodendrocytes are the glial cells responsible for myelin formation. Myelination occurs during the first postnatal weeks and, in rodents, is completed during the third week after birth. Myelin ensures the fast conduction of the nerve impulse; in the adult, myelin proteins have an inhibitory role on axon growth and regeneration after injury. During brain development, oligodendrocytes precursors originating in multiple locations along the antero-posterior axis actively proliferate and migrate to colonize the whole brain. Whether the initial interactions between oligodendrocytes and neurons might play a functional role before the onset of myelination is still not completely elucidated. In this article, we addressed this question by transgenically targeted ablation of proliferating oligodendrocytes during cerebellum development. Interestingly, we show that depletion of oligodendrocytes at postnatal day 1 (P1) profoundly affects the establishment of cerebellar circuitries. We observed an impressive deregulation in the expression of molecules involved in axon growth, guidance and synaptic plasticity. These effects were accompanied by an outstanding increase of neurofilament staining observed 4 hours after the beginning of the ablation protocol, likely dependent from sprouting of cerebellar fibers. Oligodendrocyte ablation modifies localization and function of ionotropic glutamate receptors in Purkinje neurons. These results show a novel oligodendrocyte function expressed during early postnatal brain development, where these cells participate in the formation of cerebellar circuitries, and influence its development.

Oligodendrocyte, Astrocyte, and Microglia Crosstalk in Myelin Development, Damage, and Repair.

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Oligodendrocytes are the myelinating glia of the central nervous system. Myelination of axons allows rapid saltatory conduction of nerve impulses and contributes to axonal integrity. Devastating neurological deficits caused by demyelinating diseases, such as multiple sclerosis, illustrate well the importance of the process. In this review, we focus on the positive and negative interactions between oligodendrocytes, astrocytes, and microglia during developmental myelination and remyelination. Even though many lines of evidence support a crucial role for glia crosstalk during these processes, the nature of such interactions is often neglected when designing therapeutics for repair of demyelinated lesions. Understanding the cellular and molecular mechanisms underlying glial cell communication and how they influence oligodendrocyte differentiation and myelination is fundamental to uncover novel therapeutic strategies for myelin repair.

Membrane traffic in myelinating oligodendrocytes.

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In the central nervous system (CNS), the myelin sheath is synthesised by oligodendrocytes as a specialised subdomain of an extended plasma membrane, reminiscent of the segregated membrane domains of polarised cells. Myelination takes place within a relatively short period of time and oligodendrocytes must have adapted membrane sorting and transport mechanisms to achieve such a

high rate of myelin synthesis and to maintain the unique organisation of the myelin membrane. In adult life, maintenance of the functional myelin sheath requires a carefully orchestrated balance of myelin synthesis and turnover. Imbalance in these processes may cause dys- or demyelination and disease. This review summarises what is currently known about myelin protein trafficking and mistrafficking in oligodendrocytes. We also present data demonstrating distinct transport pathways for myelin structural proteins and the expression of SNARE proteins in differentiating oligodendrocytes. Myelinating glial cells may well serve as a model system for studying general aspects of membrane trafficking and organisation of membrane domains.

Developmental increase of aspartoacylase in oligodendrocytes parallels CNS myelination.

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Canavan disease, an autosomal-recessive neurogenetic disorder, is caused by mutations in aspartoacylase, an enzyme that deacetylates N-acetylaspartate to generate free acetate in the brain. Earlier studies have shown that aspartoacylase is primarily restricted to myelin synthesizing cells (oligodendroglia) in the CNS. These findings have led us to investigate the developmental expression of aspartoacylase gene in the rat brain in an attempt to shed more light on the role of this enzyme in myelination. In situ hybridization using a 35S riboprobe based on murine aspartoacylase cDNA was used in this study. The probe hybridized mostly to the white matter tracts with different densities depending on the age of the animal and region of the brain examined. Little or no hybridization signals were detected in the 1-day-old rats, whereas the signal was clearly detectable in most of the white matter regions of the CNS in the 11-day-old rats. The signal density markedly increased at postnatal day 17, the peak of myelination. Thereafter, the hybridization signals decreased somewhat but still could be observed in the adult animals. Thus, the developmental expression pattern of aspartoacylase gene in the postnatal brain closely parallels myelination in the CNS. In the CNS, the hybridization signal of ASPA appeared to be restricted primarily to oligodendrocytes, the primary myelin synthesizing cell type in the CNS. However, the signal was not detectable in rat sciatic nerve (Schwann cells) of the peripheral nervous system. These findings indicate that the role of N-acetylaspartate in myelin synthesis is restricted to the CNS. Furthermore, they provide additional support for the acetate deficiency hypothesis of Canavan disease and also make a stronger case for acetate supplementation as an immediate and inexpensive therapy for Canavan disease.

Alterations of Oligodendrocyte and Myelin Energy Metabolism in Multiple Sclerosis.

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Multiple sclerosis (MS) is a complex autoimmune disease of the central nervous system (CNS), characterized by demyelination and neurodegeneration. Oligodendrocytes play a vital role in maintaining the integrity of myelin, the protective sheath around nerve fibres essential for efficient signal transmission. However, in MS, oligodendrocytes become dysfunctional, leading to myelin damage and axonal degeneration. Emerging evidence suggests that metabolic changes, including mitochondrial dysfunction and alterations in glucose and lipid metabolism, contribute significantly to the pathogenesis of MS. Mitochondrial dysfunction is observed in both immune cells and oligodendrocytes within the CNS

of MS patients. Impaired mitochondrial function leads to energy deficits, affecting crucial processes such as impulse transmission and axonal transport, ultimately contributing to neurodegeneration. Moreover, mitochondrial dysfunction is linked to the generation of reactive oxygen species (ROS), exacerbating myelin damage and inflammation. Altered glucose metabolism affects the energy supply required for oligodendrocyte function and myelin synthesis. Dysregulated lipid metabolism results in changes to the composition of myelin, affecting its stability and integrity. Importantly, low levels of polyunsaturated fatty acids in MS are associated with upregulated lipid metabolism and enhanced glucose catabolism. Understanding the intricate relationship between these mechanisms is crucial for developing targeted therapies to preserve myelin and promote neurological recovery in individuals with MS. Addressing these metabolic aspects may offer new insights into potential therapeutic strategies to halt disease progression and improve the quality of life for MS patients.

Expression of the neu oncogene under the transcriptional control of the myelin basic protein gene in transgenic mice: generation of transformed glial cells.

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We have taken a transgenic approach in an effort to specifically transform oligodendrocytes, the myelinating glial cells of the central nervous system (CNS). Transgenic mice were generated with a DNA construct that contained the activated neu oncogene under the transcriptional control of the myelin basic protein (MBP) gene. The MBP/c-neu transgenic animals have experienced a low incidence of brain tumors that express molecular markers specific to oligodendrocytes, providing a mouse model to study the formation and progression of oligodendrocyte tumors. A tumor from a transgenic animal has been dispersed in culture, and transformed cells that express properties of oligodendrocytes and astrocytes have been maintained. The degree to which these cells express phenotypic characteristic of oligodendrocytes or astrocytes is influenced by culture conditions. These transformed cells should serve as a valuable resource with which to study various molecular and biochemical aspects of the myelination process, as well as the lineage interrelationship of CNS glial cells.

A new player in CNS myelination.

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The formation of the myelin sheath in the CNS is the endpoint of a defined developmental program along which oligodendrocytes progress. However, the molecular signals required for the initiation of myelination are largely unknown. Ishibashi et al. report in this issue of Neuron that ATP released by axons as a result of electrical stimulation serves as an important myelination signal. Surprisingly, they found that ATP does not act directly on oligodendrocytes but rather on astrocytes, causing the release of leukemia inhibitory factor (LIF), which in turns affects promyelinating oligodendrocytes. These findings uncover a novel role for astrocytes in mediating the intricate communication between axons and myelinating glial cells.

Dynamics and mechanisms of CNS myelination.

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Vertebrate myelination is an evolutionary advancement essential for motor, sensory, and higher-order cognitive function. CNS myelin, a multilamellar differentiation of the oligodendrocyte plasma membrane, ensheaths axons to facilitate electrical conduction. Myelination is one of the most pivotal cell-cell interactions for normal brain development, involving extensive information exchange between differentiating oligodendrocytes and axons. The molecular mechanisms of myelination are discussed, along with new perspectives on oligodendrocyte plasticity and myelin remodeling of the developing and adult CNS.

Epidermal Growth Factor Pathway in the Age-Related Decline of Oligodendrocyte Regeneration.

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Oligodendrocytes (OLs) are specialized glial cells that myelinate CNS axons. OLs are generated throughout life from oligodendrocyte progenitor cells (OPCs) via a series of tightly controlled differentiation steps. Life-long myelination is essential for learning and to replace myelin lost in age-related pathologies such as Alzheimer's disease (AD) as well as white matter pathologies such as multiple sclerosis (MS). Notably, there is considerable myelin loss in the aging brain, which is accelerated in AD and underpins the failure of remyelination in secondary progressive MS. An important factor in age-related myelin loss is a marked decrease in the regenerative capacity of OPCs. In this review, we will contextualize recent advances in the key role of Epidermal Growth Factor (EGF) signaling in regulating multiple biological pathways in oligodendroglia that are dysregulated in aging.

Emerging concepts in oligodendrocyte and myelin formation, inputs from the zebrafish model.

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Oligodendrocytes (OLs) are the myelinating cells of the central nervous system (CNS), which are derived from OL precursor cells. Myelin insulates axons allowing the saltatory conduction of action potentials and also provides trophic and metabolic supports to axons. Interestingly, oligodendroglial cells have the capacity to sense neuronal activity, which regulates myelin sheath formation via the vesicular release of neurotransmitters. Neuronal activity-dependent regulation of myelination is mediated by specialized interaction between axons and oligodendroglia, involving both synaptic and extra-synaptic modes of communications. The zebrafish has provided key advantages for the study of the myelination process in the CNS. External development and transparent larval stages of this vertebrate specie combined with the existence of several transgenic reporter lines provided key advances in oligodendroglial cell biology, axo-glial interactions and CNS myelination. In this publication, we reviewed and discussed the most recent knowledge on OL development and myelin formation, with a focus on mechanisms regulating these fundamental biological processes in the zebrafish. Especially, we highlighted the critical function of axons and oligodendroglia modes of communications and calcium signaling in myelin sheath formation and growth. Finally, we reviewed the relevance of these knowledge's in demyelinating diseases and drug discovery of pharmacological compounds favoring myelin regeneration.

